Data Mining Assignment 2
Question 2:Route Blocking for Reducing Forest Fire Spread

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March 16, 2026

1 Task 1 : Optimization Formulation

Let the road network be represented as a directed graph $G = (V, E)$ where V is the set of nodes (locations) and E is the set of edges (routes). Each edge $(u, v) \in E$ has an associated propagation probability p_{uv} which represents the likelihood that fire spreads from node u to node v.

Let $R \subseteq V$ denote the set of blocked routes(edges) and let k be the maximum number of routes that can be blocked.

Decision Variables

For each edge $(u, v) \in E$, define a binary decision variable

$$x_{uv} = \begin{cases} 1 & \text{if } (u, v) \in R \\ 0 & \text{otherwise} \end{cases}$$

Objective Function

Let $\sigma(R)$ denote the expected number of nodes burnt by the fire after blocking(or removing) the edges in R from E. The goal is to minimize $\sigma(R)$.

Constraints

The maximum number of edges that can be blocked is k, i.e. $|R| \leq k$.

Formulation

$$\min_{x_{uv}} \ \sigma(R)$$

s.t.

$$\sum_{(u,v)\in E} x_{uv} \leq k$$

$$x_{uv} \in \{0, 1\}, \quad \forall (u, v) \in E$$

2 Task 2 : NP-Hardness

We prove that the Route Blocking Problem is NP-hard by providing a polynomial-time reduction from the *Influence Maximization* problem, which is known to be NP-hard under the Independent Cascade (IC) model [1],which is stated as follows:

Given a directed probabilistic graph $G = (V, E, p)$ where V is the set of nodes, E is the set of edges, and $p(u, v)$ denotes the probability that influence propagates from u to v, the goal is to choose a seed set $S \subseteq V$ with $|S| = k$ that maximizes the expected number of activated nodes:

$$\max_{|S|=k} \ \sigma(S)$$

where $\sigma(S)$ denotes the expected spread starting from seed set S.

The objective of our route blocking problem is to block at most k edges so that the expected number of nodes reached by the fire is minimized:

$$\min_{|R|=k} \sigma(R)$$

where $R \subseteq E$ is the set of blocked edges and $\sigma(R)$ denotes the expected spread after removing those edges.

Reduction Construction

Given an instance of the Influence Maximization problem on graph $G = (V, E, p)$, we construct a new graph $G' = (V', E', p')$ for the Route Blocking Problem as follows:

- Introduce a new node s acting as a super-source: V' = V $\cup\{s\}$
- For every node $v \in V$, add a directed edge (s, v): E' = E $\cup\{(s, v) \mid v \in V\}$
- Assign propagation probabilities: p'(s,v) = 1, while keeping the original probabilities unchanged.
- Set the initial burning node as $A_0 = \{s\}$.

Correctness of the Reduction

In the constructed graph, node s attempts to ignite all nodes in V. Since $p'(s, v) = 1$, node v will burn unless edge (s, v) is blocked. Blocking k edges among $\{(s, v) \mid v \in V\}$ prevents those nodes from receiving initial activation.

The nodes whose edges remain unblocked become the effective seed set and subsequently propagate influence through the network. Thus selecting which edges to block corresponds directly to selecting the seed set in the Influence Maximization problem.

Polynomial-Time Transformation

The reduction requires adding one node and $|V|$ edges while keeping the original graph structure unchanged. Therefore, the transformation runs in O($|V|$ + $|E|$) time, which is polynomial in the input size.

Since Influence Maximization is NP-hard and any instance of it can be transformed in polynomial time into an instance of the Route Blocking Problem, solving the Route Blocking Problem would solve Influence Maximization. Therefore, the Route Blocking Problem is NP-hard.

3 Task 3 : Structural Properties

Let $G = (V, E, p)$ be a directed probabilistic graph. $R \subseteq E$ denote a set of blocked edges and let $\sigma(R)$ denote the expected number of nodes reached by the fire when the edges in R are removed from the graph.

3.1 Monotonocity

A function f is monotonic if for any two sets R_1 and R_2 such that $R_1 \subseteq R_2$, we have

$$f(R_1) \leq f(R_2).$$

Claim. The function $\sigma(R)$ is not monotonic. It is infact anti-monotonic with respect to the set of blocked edges.

Proof. Let R_1 and R_2 be two sets of blocked edges such that

$$R_1 \subseteq R_2.$$

This means that R_2 blocks all edges in R_1 and possibly additional edges. Therefore, the diffusion process in the graph with blocked edges R_2 has fewer available paths for the fire to propagate compared to the graph with blocked edges R_1.Thus, the number of nodes reachable from the seed set when blocking R_2 is at most the number reachable when blocking R_1.

Therefore,

$$\sigma(R_2) \leq \sigma(R_1).$$

Hence,

$$\sigma(R_1) \geq \sigma(R_2),$$

which satisfies the definition of anti-monotonicity. Therefore, the function $\sigma(R)$ is not monotonic.

3.2 Submodularity

A set function $f : 2^E \to R$ is said to be *submodular* if for any two sets $A \subseteq B \subseteq E$ and any element $e \in E \setminus B$, the following diminishing returns property holds:

$$f(A \cup \{e\}) - f(A) \geq f(B \cup \{e\}) - f(B).$$

Claim. The function $\sigma(R)$ is not guaranteed to be submodular.
Proof. Let $R_1 \subseteq R_2 \subseteq E$ and let $e \in E \setminus R_2$. Since $R_1 \subseteq R_2$, we have

$$R_1 \cup \{e\} \subseteq R_2 \cup \{e\}.$$

By the anti-monotonicity of $\sigma(R)$,

$$\sigma(R_1 \cup \{e\}) \geq \sigma(R_2 \cup \{e\}).$$

Subtracting $\sigma(R_1)$ from both sides gives

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) \geq \sigma(R_2 \cup \{e\}) - \sigma(R_1) \tag{i}$$

Now, since $R_1 \subseteq R_2$, by anti-monotonicity we also have

$$\sigma(R_2) \leq \sigma(R_1).$$

Thus,

$$\sigma(R_2 \cup \{e\}) - \sigma(R_2) \geq \sigma(R_2 \cup \{e\}) - \sigma(R_1).$$

Rearranging,

$$\sigma(R_2 \cup \{e\}) - \sigma(R_1) \leq \sigma(R_2 \cup \{e\}) - \sigma(R_2) \tag{ii}$$

From inequalities (i) and (ii), we obtain

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) \geq \sigma(R_2 \cup \{e\}) - \sigma(R_1) \leq \sigma(R_2 \cup \{e\}) - \sigma(R_2).$$

From the above inequalities, we cannot always infer

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) \geq \sigma(R_2 \cup \{e\}) - \sigma(R_2).$$

Hence, the function $\sigma(R)$ does not necessarily satisfy the diminishing returns property. Therefore, $\sigma(R)$ is not guaranteed to be submodular.

4 Task 4: Algorithm for Route Blocking

We propose a greedy route-blocking algorithm using the **Cost-Effective Lazy Forward (CELF)** optimization. To improve scalability, we introduce a **dynamic pruning step** that restricts the candidate edges to those within a limited BFS depth from the seed set in sampled graph realizations.

The intuition is that edges far from the initial fire source rarely influence the final spread, so evaluating them is unnecessary.

4.1 Algorithm

Algorithm 1 CELF Route Blocking with Dynamic Pruning

Require: Graph $G(V, E)$, seed set S, budget k, simulations r, hop limit H
Ensure: Blocked edge set B

```
Generate r random realizations {G_1, ..., G_r}
d_limit ← max(3, ⌊(max BFS depth from S)/3⌋)
C ← ∅
for each realization G_i do
    Run BFS from S up to depth H
    Add edges at depth ≤ d_limit to candidate set C
end for
B ← ∅
σ_base ← EstimateSpread(S, B)
Initialize max-heap Q
for each e ∈ C do
    Δ_e ← σ_base − EstimateSpread(S, B ∪ {e})
    Push (e, Δ_e, 0) into Q
end for
for i = 1 to k do
    while true do
        Pop (e, Δ_e, iter) from Q
        if iter = i then
            B ← B ∪ {e}
            σ_base ← σ_base − Δ_e
            break
        end if
        Δ_new ← σ_base − EstimateSpread(S, B ∪ {e})
        Push (e, Δ_new, i) into Q
    end while
end for
return B
```

4.2 Derivation of Computational Complexity

Let $n = |V|$, $m = |E|$, r be the number of Monte Carlo realizations, and $|C|$ be the number of candidate edges after pruning ($|C| \ll m$).

- Candidate generation: $O(r(n + m))$
- CELF initialization: $O(|C|\, r(n + m))$
- Greedy selection (worst case): $O(k\, |C|\, r(n + m))$

Thus the overall worst-case complexity is

$$O\big(k\, |C|\, r(n + m)\big).$$

In practice, CELF significantly reduces recomputation of marginal gains, making the algorithm much faster than the worst-case bound.

Approximation Guarantee

The proposed algorithm does **not** provide a constant-factor approximation guarantee.

Greedy guarantees such as $(1 - 1/e)$ require the objective function to be **submodular**. However, the wildfire containment objective does not satisfy the diminishing returns property.

Consider a seed node connected to a large cluster X via two edges e_1 and e_2. Blocking either edge individually does not prevent the fire from reaching X, but blocking both prevents the entire cluster from burning. Let $S = \emptyset$ and $T = \{e_1\}$:

$$f(\{e_2\}) - f(\emptyset) = 0$$

$$f(\{e_1, e_2\}) - f(\{e_1\}) = |X|$$

Since $0 \not\geq |X|$, the diminishing returns property fails. Therefore the objective is not submodular, and a bounded approximation guarantee does not hold.

5 Task 5 : Implementation

We list down the results in tables in two different scenarios below:

Table 1: Results with Unlimited Hops

Metric	Value
$\sigma(\emptyset)$	36.2800
$\sigma(R)$	8.2200
Edges Evaluated ($\|R\|$)	50
Budget (k)	50
Reduction	28.0600
Reduction (%)	77.34%
$\rho(R)$	0.773429
Number of Simulations	50
Hops	Unlimited

Table 2: Results with 3-Hop Constraint

Metric	Value
$\sigma(\emptyset)$	124.4200
$\sigma(R)$	10.0000
Edges Evaluated ($\|R\|$)	50
Budget (k)	50
Reduction	114.4200
Reduction (%)	91.96%
$\rho(R)$	0.919627
Number of Simulations	50
Hops	3

References

[1] D. Kempe, J. Kleinberg, and É. Tardos, "Maximizing the spread of influence through a social network," in *Proceedings of the 9th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2003, pp. 137–146.

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[3] H. Tong, B. A. Prakash, C. Tsourakakis, M. Faloutsos, and C. Faloutsos, "On the Vulnerability of Networks," in *10th IEEE International Conference on Data Mining (ICDM)*, 2010, pp. 511-520.

[4] A. Krause and D. Golovin, "Submodular Function Maximization," *Tractability: Practical Approaches to Hard Problems*, Cambridge University Press, 2014.

[5] OpenAI, "ChatGPT," Available: https://chat.openai.com/,